DERIVATIVES

From Rolle's Theorem to the Mean Value Theorem

The Mean Value Theorem is primarily a theoretical tool.

Rolle's Theorem

$$f(a) = f(b) \implies f'(c) = 0$$

SPECIAL CASE
$f(a) = f(b)$

Mean Value Theorem

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

PROVES

Uses

- A function with positive derivative on an interval is increasing
- Two functions with the same derivative differ by a constant
- The connection between antiderivatives and definite integrals in the Fundamental Theorem of Calculus